FTD Master

AI-Powered ESG Analysis
Session 1 – Course overview, Intro

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UNIVERSITÉ PARIS 1
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About Your Instructor

Rami El Mnebhi is Head of Product at DINUM/Etalab, leading Albert API — France's sovereign LLM platform deployed on SecNumCloud infrastructure. He holds an engineering degree from INSA Lyon and has 10+ years of experience in data engineering, ML, and LLM/RAG systems across Société Générale, BNP Paribas CIB, and now the French public sector. He built and shipped production RAG platforms at scale and is passionate about open-source AI and public-interest technology.

Course Overview

- 6 sessions from April 7th to May 26th
- **Objective**: **Build a functional ESG Assistant powered by LLMs**
- Hands-on approach: Lectures, coding, group work
- Final team project presented in session 6
- Best individual contributor wins:

  → *Sustainability and Climate Risk* Certification

1 — Apr 7 — Intro & First LLM Demo

2 — Apr 14 — RAG: Theory + HyDE + MQR

3 — May 5 — [Dev Session]

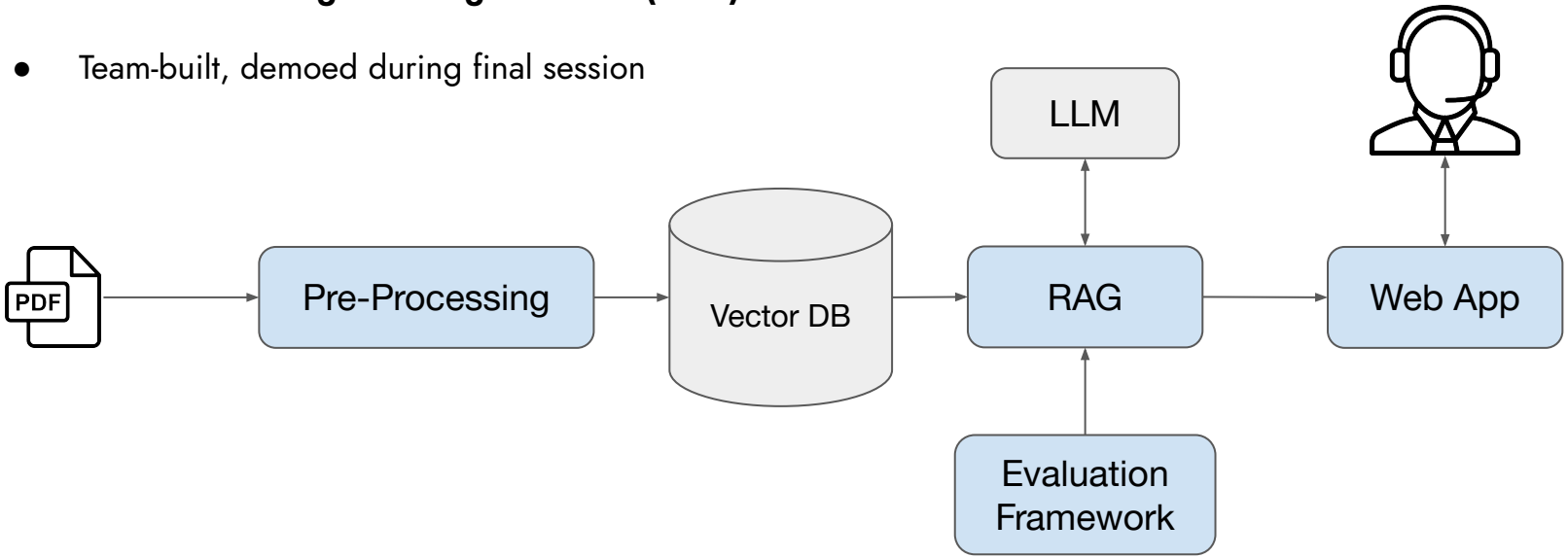
4 — May 12 — Advanced RAG + MCP

5 — May 19 — [Dev Session]

6 — May 26 — Demo Day

What You Will Build

- A **web-based ESG Assistant** powered by LLMs
- Ability to analyze **ESG-related PDFs (and acquire them)**
- **Answer natural language questions with citations**
- Use **retrieval-augmented generation (RAG)** architecture
- Team-built, demoed during final session



Tools & Tech Stack

🛠️ Languages & Frameworks

- Python
- Jupyter Notebooks
- Streamlit or FastAPI (or anything you chose)
- Continue.dev / OpenCode coding assistant

🧠 AI & LLM APIs

- Albert API (sovereign French LLM — OpenAI-compatible)
- OpenAI GPT-4o (fallback / comparison)

📄 Document Handling

- PyMuPDF or pdfplumber (PDF parsing)
- LangChain / LlamaIndex / LangGraph (agents)

📦 Vector Database

- Qdrant/ ChromaDB / PostgreSQL (to store document chunks)

📊 Evaluation Tools

- RAGAS,DeepEval (for output quality measurement)

Weekly Progression

Session	Date	Focus
1️⃣	Apr 7	Intro, ESG Context,DataSet,First LLM Call
2️⃣	Apr 14	RAG Theory, HyDE & Multi-Query Retrieval
3️⃣	May 5	Dev Session (guided build time)
4️⃣	May 12	Advanced RAG, Reranking, MCP & RAGAS
5️⃣	May 19	Dev Session (guided build time)
6️⃣	May 26	Demo Day

Group Project Rules

- Work in **teams** of 3 students (8 teams of 3)
- Each team builds **one ESG Assistant**
- You'll receive:
    - A shared **API key** (Albert API)
    - A set of **ESG-related PDF and dataset**
    - A question dataset
- Use your budget wisely: **LLM calls are limited** (overall cost will count in final evaluation :))
- Code assistance (e.g., via Continue) allowed, but counts toward usage
- Collaboration encouraged — **plagiarism strictly forbidden** but usage of AI is recommended
- Best student receives a free GARP Sustainability & Climate Risk (SCR) certification

Evaluation Criteria

👥 Group Project (70%)

- ✅ Functionality — 25%
- 📚 Answer Relevance & Quality — 20%
- 🧠 Technical Implementation — 10%
- 🎨 UI/UX — 5%
- 🌟 Bonus: Creativity / Originality — +5%

🙋 Individual Contribution (30%)

- 🗳️ Peer Voting (Most Valuable Teammate) — 15%
- 👀 Instructor Observation + attendance — 10%
- ⚙️ Optional Challenge (e.g. best prompt, evaluation logic, advanced LLM usage) — +5%

All Teams Present their demos on Session 6 → Instructor Selection of Best Team → Peer Voting of Best Student → Best Student

Beyond the Project

In addition to your team project, each of you can choose an area of focus to explore more deeply throughout the course.

These are topics that go beyond the mandatory steps and help you grow toward industry-ready skills.

🔍 **Pick what excites you the most:**

- 🔧 **Prompt Engineering**: master system prompts, templates, and tricks
- 📚 **Advanced NLP for ESG Docs**: table parsing, entity extraction, hybrid retrieval
- 🧱 **App Deployment & Architecture**: move from notebook to real-time app
- 🤖 **Coding with AI Assistants**: integrate Continue, test pair programming with LLMs
- ⚙️ **LLM Ops**: how to host, fine-tune, or serve LLMs on-prem or in the cloud
- 🧠 **Your Own Proposal** (pitch it!)

But First .... The Use Case

CSRD: The Regulation Driving ESG Disclosure

- The **Corporate Sustainability Reporting Directive** (EU, 2024) requires ~50,000 European companies to publish standardized sustainability data annually
- Replaces the older NFRD — much broader scope, much stricter standards
- **Built around double materiality: companies report both how ESG risks affect them financially, and how their activities affect people and planet**
- Structured around **ESRS (European Sustainability Reporting Standards)** — 12 thematic standards covering climate, biodiversity, social, governance
- Analysts at banks, asset managers, and rating agencies must read hundreds of these reports per year — the bottleneck is extracting the right answer from the right page of a 300-page report

Who Uses This Assistant and How?

- Emma, **ESG Analyst at a French asset manager**. She is evaluating TotalEnergies for fund inclusion and needs to verify 6 ESRS datapoints before tomorrow's investment committee.

- Without the assistant: download a 280-page report, Ctrl+F across 3 documents — **4 hours of work, high risk of missing context**

- **With the assistant — she types:**

  "What are TotalEnergies' Scope 1 and 2 GHG emissions for 2023 and their 2030 reduction target?"

  "Does TotalEnergies report against ESRS E1? What climate transition plan do they disclose?"

  "What percentage of TotalEnergies' CapEx is aligned with the EU Taxonomy?"

- These are real CSRD analyst questions. By the end of this course, your assistant answers them with citations grounded in the actual report.

Where Does ESG Data Come From?

- ESG assistants are only as good as the documents they retrieve from. **Before building a RAG pipeline, you need a corpus.**

- Corporate sustainability reports are published annually as PDFs on company IR pages — **but there is no central repository. You have to go get them.**

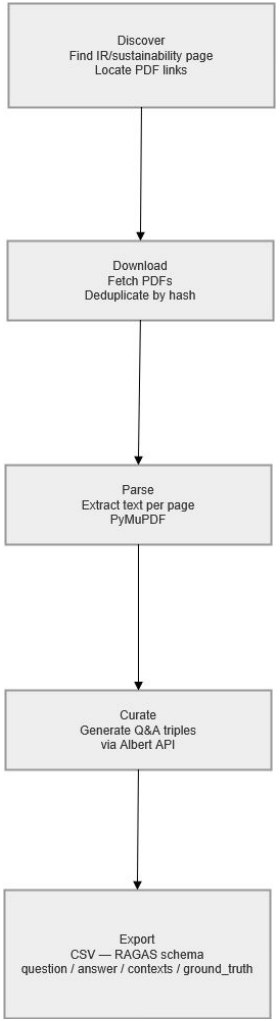
- **What we build today: a scraper that finds, downloads, and parses ESG reports from major CSRD-reporting companies, then generates a structured Q&A dataset to use as our RAG evaluation benchmark in Session 4.**

- Target sources: TotalEnergies, BNP Paribas, Airbus, Danone, Engie — **all publish CSRD-aligned reports with consistent structure**

- Output: curated_esg_dataset.csv + evaluation questions ready for evaluation pipeline

The Scraping Pipeline

- Step 1 — **Discover**: find the sustainability/IR page, locate PDF links (requests + BeautifulSoup or Crawl4AI for JS-heavy sites)
- Step 2 — **Download: fetch and store PDFs locally, deduplicate by hash**
- Step 3 — **Parse**: extract clean text per page with PyMuPDF, structure by section when possible
- Step 4 — Curate: generate question/answer/context triples using Albert API, export to CSV matching RAGAS schema (question, answer, contexts, ground_truth)



Before You Scrape: The Legal Reality

- Generally fine: **publicly published PDFs linked from official IR/sustainability pages**
- Respecting crawl delays, no rate abuse, single downloads
- **Academic and educational use (which this is)**
- Data companies are legally required to publish under CSRD **Not fine: ignoring robots.txt, scraping behind login walls, bulk crawling without throttling, redistributing raw PDFs**
- Always check robots.txt before scraping. In production, prefer official data providers (Bloomberg ESG, MSCI, Sustainalytics). For this course we scrape only public documents for educational purposes.

Ressources

- Class GitHub Repo : https://github.com/RamiELM/FTD_2026_ESG_AI_Assistant

- IDE : https://code.visualstudio.com/download

- Coding assistance : https://docs.continue.dev/getting-started/overview
- Vibe Coding : https://opencode.ai/

- Albert API : https://albert.api.etalab.gouv.fr https://doc.incubateur.net/alliance/albert-api/prise-en-main

- RAG Framework :
  - https://www.langchain.com/
  - https://www.llamaindex.ai/

- Sample Dataset : https://www.kaggle.com/datasets/jaidityachopra/esg-sustainability-reports-of-s-and-p-500-companies

- WebUI :
  - https://streamlit.io/
  - https://www.gradio.app/
  - https://openwebui.com/

- RAG Evaluation : https://docs.ragas.io/en/stable/ https://github.com/confident-ai/deepeval